



CLINICAL GUIDELINE

Adnexal torsion

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TARGET AUDIENCE and SETTING

Monash Health medical staff.

In the setting of female patients presenting with signs and symptoms of adnexal torsion.

STANDARD REQUIREMENTS

When undertaking any clinical interaction with a patient, staff are expected to:

- Perform routine hand hygiene. Refer to the [Hand Hygiene Procedure](#).
- Introduce yourself to the Patient and Carer/ Family if in attendance.
- Explain what you do and what you are there for
- Check patient identification. Refer to the [Patient Identification Procedure](#).
- Obtain consent as per the [Consent to Medical Treatment Procedure](#).
- Keep the patient/carers informed and involve them in decision making.
- Document interaction in the Electronic Medical Record or health record using black pen; including date, time, signature and designation.

CLINICAL GUIDELINE

1. Introduction

Adnexal torsion is one of the most common gynaecological surgical emergencies.¹ It is defined as partial or complete rotation of an adnexum on its vascular pedicle. Of particular concern, it is most likely to occur in children, adolescents, and those of reproductive age (average age 26).² Given the potential loss of ovarian function with prolonged torsion, it is important to make an early diagnosis and proceed with management in a timely fashion.

Isolated fallopian tube torsion is uncommon in any age group, but may also occur and adversely impact future tubal function.³ Tubal torsion may occur either in the mid-portion of the tube or around its ligamentous supports.

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Ovarian torsion may result in impedance of blood supply. Venous blood flow is the first to be impaired, followed by arterial flow. This leads to congestion, adnexal oedema, ischemia, and ultimately necrosis. The duration of ischemia required for irreversible damage is unknown.

2. Risk factors

Ovarian mass: An ovarian mass (physiologic cyst or neoplasm) is the primary risk factor for ovarian torsion, with an incidence of 85-95%.⁴

Size: In general, torsion is most likely to occur when the ovary is 5 cm in diameter or larger. In three case series, each with 80 or more patients, 83 to 93% of torsed ovaries were 5 cm or larger.⁵ However, torsion may occur with a mass of any size. Although it has been postulated that very large ovaries are less likely to undergo torsion because the size impedes movement, there is no size at which torsion becomes impossible.

Benign versus malignant masses: Torsion is more likely to occur with benign cysts or neoplasms rather than malignant lesions, possibly because malignant masses are more likely to be non-mobile. Case series have reported malignancy in 2% or fewer of cases of torsion.⁶ However, the rate of malignancy in a torsed ovary is higher among postmenopausal women.⁷

Normal ovaries: Importantly normal ovaries can undergo torsion particularly in paediatric patients.⁸ In children under 15 years, normal ovaries have been demonstrated in over 50% of patients with ovarian torsion.⁹

Reproductive age: The majority of cases of ovarian torsion occur in women of reproductive age⁶ but there is no upper or lower age limit.

Pregnancy: Pregnancy is associated with an increased risk of ovarian torsion, accounting for 10 to 22 percent of cases of torsion.¹⁰

Ovulation induction: Ovulation induction increases risk due to ovarian enlargement and may occur in up to 8% of women with Ovarian hyperstimulation syndrome (OHSS).¹¹

Prior adnexal torsion: Incidence unknown. Up to 11% reported as recurrent in ipsilateral or contralateral ovary.¹⁰ It must be noted that the recurrent torsion can occur in the same "salvaged ovary" or can be asynchronous bilateral ovarian torsion.¹²

3. Clinical presentation

The classic presentation of ovarian torsion is acute onset moderate to severe unilateral pelvic pain, often with nausea and vomiting, in the presence of an adnexal mass.^{13,14} However, the presentation may vary.

Symptoms and signs are non-specific and torsion may also occur in the absence of an adnexal mass. Thus, a high index of suspicion is required, and in any female presenting with acute onset unilateral abdomino-pelvic pain, torsion is the diagnosis until proven otherwise.

Fever (usually low grade) may be a marker of adnexal necrosis, particularly in the setting of leukocytosis¹⁵ and warrants urgent laparoscopy.

4. Evaluation

Medical history

Ask specifically about nausea, pain and vomiting, the cardinal features of torsion. There may be a history, or recent diagnosis, of an adnexal mass. Recent vigorous physical activity may precede torsion and will increase suspicion in equivocal cases.

Physical examination

A low grade fever, elevated blood pressure and mild tachycardia may be present. An abdominal and bimanual examination must be performed but again signs are non-specific and variable.

Tenderness on examination is absent in as many as one-third of patients.⁴ A palpable pelvic mass and localized pain may or may not be present. Peritonism suggests adnexal necrosis.⁵

Laboratory evaluation

Request full blood examination (FBE), bhCG, urea & electrolytes (U&E) Creatinine (Cr).

Group & hold (G&H) is not indicated.

Leukocytosis is present in 20% to 56% of cases.¹⁶ This finding is nonspecific and may not be present on initial presentation. C-reactive protein (CRP) may be elevated and is a non-specific marker for necrosis.¹⁷

Imaging

Ultrasound with colour flow Doppler is the imaging modality of choice for suspected adnexal torsion.¹⁸ The most frequent findings on ultrasound are ovarian enlargement and the absence of Doppler flow.¹⁹ Absent blood flow is consistent with torsion. However, normal flow may be seen if the ovary has transiently detorted or is partially tortored, or if the ultrasound is performed early in the torsion process when perfusion is preserved and only venous and lymphatic drainage are obstructed.¹⁸ The ovary has a dual blood supply therefore flow may be preserved in the presence of torsion. Other sonographic findings that are associated with ovarian torsion are:

- An ovary that is rounded and enlarged compared with the contralateral ovary (due to oedema and engorgement)
- An ovarian mass with probe tenderness.
- Heterogeneous appearance of the ovarian stroma – This is due to oedema and haemorrhage.
- Multiple small peripheral follicles ("string of pearls") – This is thought to be due to oedema. This may be seen in polycystic ovaries (PCO), although in the true polycystic ovary the stroma is centrally echogenic, does not appear oedematous, and pain is absent.
- Abnormal ovarian location – Ovaries usually lie lateral to the uterus, but in torsion, they may be located anterior to the uterus.

- Doppler grayscale "whirlpool sign" in the ovarian vessels – This is visualized as a round hyperechoic structure with concentric hypoechoic stripes or a tubular structure with internal heterogeneous echoes, representing twisting of the vascular pedicle. Two small series of 20 patients with suspected torsion found this sign to have a sensitivity of 90 percent or higher for the diagnosis of torsion.²⁰

MRI and CT scans may reliably identify adnexal torsion. Many women with adnexal torsion present with non-specific abdominal/pelvic pain and initially undergo CT scanning. A well reported CT scan has a reasonable negative predictive value; any positive findings or poorly described ovarian anatomy requires further evaluation.²¹ Intra-adnexal CT features of adnexal torsion included fallopian tube thickening (74%), eccentric or concentric wall thickening (54%), and eccentric septal thickening (50%). CT has a low overall sensitivity and is not the first line of imaging for suspected torsion.

MRI may be helpful if findings on ultrasound are equivocal, but the cost and time required for MRI does not justify its routine use for the evaluation of suspected adnexal torsion. MRI findings consistent with ovarian torsion include an enlarged, oedematous ovary in an abnormal location, increased free fluid and, with contrast enhancement, coiled ovarian vessels exhibiting the "whirlpool" sign.

5. Treatment

The gold standard for treatment and diagnosis in suspected adnexal torsion is laparoscopy.

Adnexal torsion remains one of few gynaecological emergencies because of its organ threatening nature. If ovarian torsion is suspected, surgical treatment must be expedited. A loss of pain does not signify resolution, rather conversely can reflect neural necrosis within the affected structures.

Laparoscopic diagnosis and treatment must be undertaken as soon as the diagnosis is suspected, ideally within 12 hours of presentation. Precise determination of the time of devascularization is impossible, and many will undergo intermittent detorsion providing some arterial perfusion, so while time is critical in the preservation of the organs there is no time that is too late for intervention.

Laparoscopy is superior to laparotomy with shorter hospital stays, less febrile morbidity and lower analgesic requirements.²²

The choice of detorsion versus removal of the structure must be strongly weighted towards detorsion, as most adnexa will re perfuse and prove to be viable.²³ Concerns about venous thromboembolism and undiagnosed malignancy have not been validated.²⁴

There is no accurate non-invasive test for tissue necrosis in this context. Laboratory markers have low sensitivity and specificity. Sensitivity and specificity of CRP for necrosis were 35% and 83%, respectively, and positive and negative predictive values were 38% and 82%.¹⁷ MRI findings in women who underwent either cystectomy or radical adnexectomy for visually confirmed ovarian torsion more consistently show haemorrhagic infarction in the adnexectomy cohort²⁵, however MRI is not a reliable triage tool.

5.1 Ovarian viability after surgical management

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In 102 cases treated with laparoscopic detorsion, 91% of those treated with laparoscopic detorsion within 16 hours (2-144) of presentation maintained ovarian function and post-surgical ovarian size and follicular development was within normal range. The median time from symptom onset to surgery was 16 hours, with a range of 2-144 hours. Most patients (91.3%) had normal findings.²³

6. Role of oophoropexy

Patients who have had ovarian torsion with a normal adnexum are at increased risk of both ipsilateral and contralateral adnexal torsion.^{26,27} The ovary has been variably fixed to the round ligament, the uterosacral ligament, the pelvic side wall or the posterior uterus.²⁸ Plication of the ovarian ligament may also be undertaken.²⁹ In the absence of high-quality studies, consider oophoropexy in paediatric patients with normally sized ovaries and adults with recurrent torsion.

7. Special populations

Pregnancy: adnexal torsion is more common in pregnancy, particularly after ovulation induction.¹¹ Laparoscopy is safe in pregnancy in experienced hands.³⁰ Curettage is contraindicated in the presence of a viable pregnancy and left lateral tilt and lower operating pressures may be considered after the second trimester.

If the corpus luteum is disrupted progesterone supplementation (200 mg PV daily) is given until 12 weeks' gestation.

Children and adolescents: Children are less likely to have an ovarian mass at the time of ovarian torsion. The evidence overwhelmingly supports detorsion only as the gross appearance of the ovary does not correlate with long term ovarian viability or function.²⁴

Post-menopausal women: Ovaries have endocrine function, supporting ovarian retention until age 65. Older women are at higher risk of malignancy and visually abnormal ovaries may be malignant. Care must be individualized balancing the risks of oophorectomy versus ovarian retention.

8. Conclusion

Adnexal torsion is a problem requiring a high index of suspicion. Clinical and ultrasound findings vary and may be unreliable.

Laparoscopy must be undertaken within 12 hours of presentation to the emergency department, and for many de torsion will be the treatment of choice. Consider Oophoropexy.

A proactive policy of laparoscopy will mean the occasional normal laparoscopy, notwithstanding other pathology which may be found. The aim is to achieve a close to zero ovarian loss rate.

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